NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 07-05-26**
Time: 3 Hrs **Max. Marks: 180**

07-05-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-12(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

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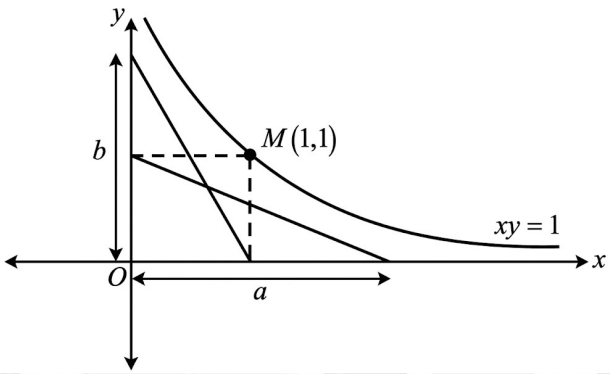
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MATHEMATICS **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. With reference to the graph shown, the value of $\tan^{-1}a+\tan^{-1}b-\tan^{-1}\left(\frac{a+b}{1-ab}\right)$ is equal to



A) 0 B) $-\pi$ C) π D) Does not exist

2. Number of points of discontinuity of the function $f(x)=\text{sgn}\left(x^2-x+1\right)+\left[x^2-x+1\right]$ in $x\in[0,3]$ ([.] denotes greatest integer function, sgn {.} denotes signum function)

A) 8 B) 5 C) 2 D) 6

3. The value of integral $\int_{-1}^{1}\frac{dx}{\left(x^2+x+1\right)+\sqrt{x^4+3x^2+1}}$ is

A) $\frac{1}{2\sqrt{2}}$ B) 0 C) $\frac{\pi}{2}$ D) $\frac{\pi}{4}$

4. L_1 is a tangent drawn to the curve $x^2-4y^2=16$ at $\left(5,\frac{3}{2}\right)$. L_2 is another tangent parallel to L_1,which meets the curve at B . L_3 and L_4 are normals to the curve at A and B . Lines L_1, L_2, L_3, L_4 forms a rectangle, then which of the following is INCORRECT?

A) Equation of tangent at B is $6y=5x+6$

B) Equation of normal at B is $12x+10y+75=0$

C) Radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$

D) Radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$

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SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. Let $(a-1)\left(x^2+\sqrt{3}x+1\right)^2-(a+1)\left(x^4-x^2+1\right)\le 0;\forall x\in R$, then which of the following is/are CORRECT?

A) $a\in\left[-\frac{2}{\sqrt{3}},\frac{2}{\sqrt{3}}\right]$

B) Largest possible value of a is $\frac{2}{\sqrt{3}}$

C) Number of possible integral values of a is 3

D) Sum of all possible integral values of a is 0

6. For the set of given equations/inequations $(log_2 x)(log_y 2)+1=0;\ sinx\,cosy=1-cosx\,siny$ and $x+y<8$, which of the following statements is/are CORRECT?

A) Number of real pairs in the form of pair (x,y) satisfying the equations/inequations is 2 .

B) Number of real pairs in the form of pair (x,y) satisfying the equations/inequations is 4.

C) Sum of real values of x satisfying the equations / inequations is the fundamental period of the function $f(x)=sin\left(\frac{4x}{5}\right)$

D) Sum of real values of y satisfying the equations/inequations is the fundamental period of $f(x)=sin\frac{4x}{5}$.

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7. Let T be the line passing through the points $P(-2,7)$ and $Q(2,-5)$. Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangent to S_1 at P and tangent to S_2 at Q , and also that S_1 and S_2 touch each other at a point, say M . Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1,1)$ be F_2. Let E_2 be the set of the midpoints of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE?

A) The point $(2,-5)$ does NOT lies in E_1

B) The point $\left(\frac{4}{5},\frac{7}{5}\right)$ does NOT lies in E_2

C) The point $\left(\frac{1}{2},1\right)$ does NOT lies in E_2

D) The point $\left(0,\frac{3}{2}\right)$ does NOT lies in E_1

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. If $L=2\sum_{n=1}^{\infty}2^n \sin^2\left(\frac{\pi}{2^{n+1}}\right)\tan\left(\frac{\pi}{2^{n+2}}\right)$, then [2L] is equal to ([.] denotes greatest integer function)

9. Let the mean and the variance of 5 observations x_1,x_2,x_3,x_4,x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observations are $\frac{7}{2}$ and 'a' respectively, then $(4a+x_5)-7$ is equal to_____

10. Assuming all the terms of A.P$[a_n]$ are integers with $a_1=2019$ and for all $n\in I^+$ there always exists positive integer m such that $a_1+a_2+a_3+.....+a_{n-2}+a_{n-1}+a_n=a_m$, then the number of such sequences is ____

11. If A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron, and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of tetrahedron is 5 cubic units, then the minimum value of $\left|\frac{(A_1+A_2+A_3+A_4)(h_1+h_2+h_3+h_4)}{120}\right|$ is____

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12. There is exactly one real $x \in \left(0, \frac{\pi}{2}\right)$ such that $\left(tan^2 \frac{x}{2}\right)\left(cot^4 x + 1\right)\left(cosec^2 x + tan^2 x\right) = 1$.

Find the positive integer k such that $cos^{2022} x = sin^k x$

13. The value of $\left[\pi \cdot \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \frac{8}{7} \cdot \frac{8}{9} \ldots\ldots \infty\right]$ is ____ . Where [.] denotes greatest integer function

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

14. Match the following List-I and List-II.

	List-I		**List-II**
P)	If A and B are non-singular real matrices such that $\det(A) = 3$ and $A^{-1} + B^{-1} = (A+B)^{-1}$, then $\det(B) =$	1)	190
Q)	Let $A = \begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $(a,b,c,x,y,z,p,q,r \in N)$. If $tr\left(AB + AB^3 + AB^5 + \ldots + AB^{19}\right) = 210$, then the ordered triplets (p,q,r) is_____(tr(A) denotes sum of principal diagonal elements of A)	2)	3
R)	Consider $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 0 & 0 \\ 5 & 0 & 0 \end{bmatrix}$. If $C = \left[C_{ij}\right]_{3\times 3} = A^{20} + B$ and $C_{22}C_{33} - C_{23}C_{32} = 2^m$, then $m =$	3)	5

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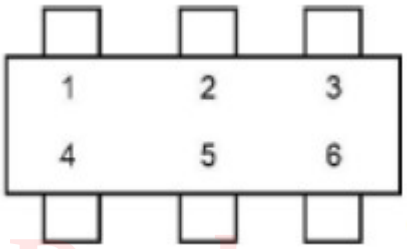
S)	Consider three matrices $A=\left[a_{ij}\right]_{3\times5}$, $B=\left[b_{ij}\right]_{5\times9}$ and $C=A\times B=\left[c_{ij}\right]_{3\times9}$. It is given that $a_{ij}=\frac{i}{3j-2}$ and $b_{ij}=\frac{j}{3i+2}$, then the value of C_{28} is_____	4)	11
		5)	20

Then CORRECT options is

A) $P\rightarrow3;Q\rightarrow4;R\rightarrow2;S\rightarrow3$ B) $P\rightarrow2;Q\rightarrow1;R\rightarrow5;S\rightarrow3$

C) $P\rightarrow3;Q\rightarrow2;R\rightarrow5;S\rightarrow1$ D) $P\rightarrow1;Q\rightarrow5;R\rightarrow3;S\rightarrow4$

15. There is a rectangular table as shown in the figure, which has three chairs on two sides each.



Three married couples sit on these chairs. Then answer the following grid.

	List-I		**List-II**
P)	Probability that couples sit either infront of each other or adjacently.	1)	0.2333
Q)	Probability that no couple sit either infront of each other or adjacently.	2)	0.1000
R)	Probability that exactly one couple sit infront of each other or adjacently.	3)	0.1555
S)	Probability that no two ladies sit infront of each other or adjacently.	4)	0.1322
		5)	0.1333

Then CORRECT options is

A) $P\rightarrow1;Q\rightarrow2;R\rightarrow3;S\rightarrow4$ B) $P\rightarrow5;Q\rightarrow3;R\rightarrow1;S\rightarrow2$

C) $P\rightarrow3;Q\rightarrow5;R\rightarrow1;S\rightarrow2$ D) $P\rightarrow3;Q\rightarrow5;R\rightarrow2;S\rightarrow1$

For More Join: @IITJEE_Advanced

16. Consider the curves $2|z-2a|=z+\bar{z}$ and $|z-4a|=3a$ where $z=x+iy$ and $a\in R^+$.

	List-I		List-II
P)	If z_1 is the point of intersection of common tangents of two curves (not parallel to imaginary axis), then $\frac{\|z_1\|}{a}$ is	1)	A prime number
Q)	If A is the area of triangle formed by chord of contact and common tangents drawn from z_1 (used in 1st part) to the curve $\|z-4a\|=3a$, then $\frac{4A}{\sqrt{3}a^2}$ is	2)	Perfect square
R)	Number of common points in both curves is	3)	Perfect cube
S)	If equation of chord of contact from z_1 (used in part I) to the curve $2\|z-2a\|=z+\bar{z}$ is $\frac{z+\bar{z}}{2}=ka$, then k is	4)	Multiple of 10
		5)	Multiple of 5

Then CORRECT options is

A) $P\to 1;Q\to 3;R\to 5;S\to 2$ B) $P\to 1;Q\to 2;R\to 5;S\to 3$

C) $P\to 1;Q\to 3;R\to 1;S\to 2$ D) $P\to 1;Q\to 2;R\to 1;S\to 3$

17. Consider $I_1=\int_0^1\frac{e^{\tan^{-1}x}}{\sqrt{1+x^2}}dx$, $I_2=\int_0^1\frac{xe^{\tan^{-1}x}}{\sqrt{1+x^2}}dx$, $I_3=\int_0^1\frac{x^2e^{\tan^{-1}x}}{\sqrt{1+x^2}}dx$, then

	List-I		List-II
P)	$I_1+I_2=$	1)	$\frac{3}{2}$
Q)	$I_1+I_2-4I_3=$	2)	1
R)	$\frac{I_1+I_2+1}{I_1+I_2+2I_3}=$	3)	$\sqrt{2}e^{\pi/4}-1$
S)	$\frac{2(I_1+I_2)}{2I_3+1}=$	4)	$\sqrt{2}e^{\pi/4}$
		5)	$\sqrt{2}e^{\pi/4}-3$

Then CORRECT options is

A) $P\to 3;Q\to 5;R\to 2;S\to 3$ B) $P\to 4;Q\to 5;R\to 1;S\to 3$

C) $P\to 3;Q\to 4;R\to 5;S\to 1$ D) $P\to 1;Q\to 5;R\to 3;S\to 4$

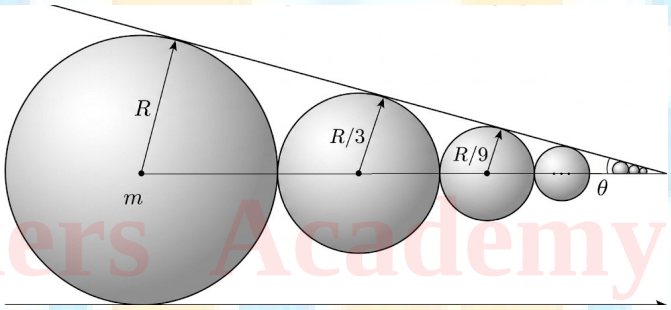
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PHYSICS **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

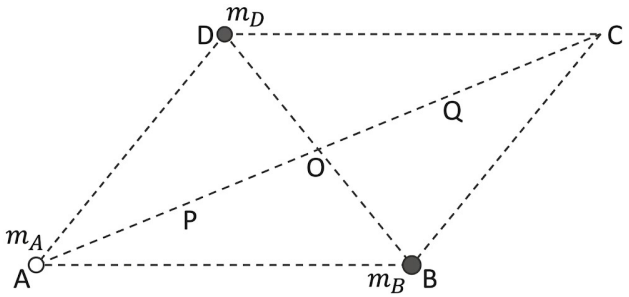
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

18. An infinite number of uniform discs having same thickness and made of same material are pivoted at their respective centres and arranged in such a manner that centres of all the discs are at the same horizontal level. Radius of biggest disc is R and successive discs have radii $\left(\frac{1}{3}\right)^{rd}$ of radius of disc to its left. A rod having same mass as the mass of largest disc is placed over the discs, touching all of them. There is no friction between the discs but friction is sufficient between the rod and the discs, so that the rod does not slip. Then the acceleration of the rod is



A) $\frac{8\,g}{25}$ B) $\frac{3\,g}{25}$ C) $\frac{g}{25}$ D) $\frac{g}{50}$

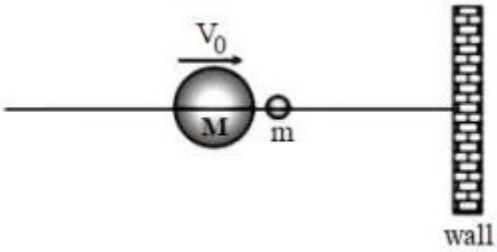
19. Three particles of masses m_A, m_B, and m_D are kept at the vertices A, B, and D respectively of a parallelogram as shown. The centre of mass of the three particles lies at the point ' P ' which is the midpoint of AO (O is the intersection point of the diagonals of the parallelogram). Now, another mass m_C has to be kept at ' C ' so that the centre of mass of the four particles lies at Q (Q is the midpoint of OC). Choose the correct option:



A) $m_C = 3\,m_A$ B) $m_C = 4\,m_A$ C) $m_C = 4\,m_B$ D) $m_C = 2\,m_D$

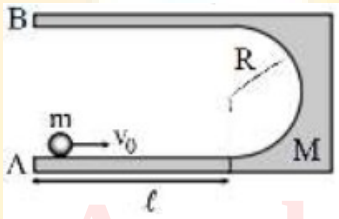
For More Join: @IITJEE_Advanced

20. Two beads of mass M and m can move without friction on a horizontal wire as shown in the figure. Initially M is moving with a speed v_0 and m is stationary. During subsequent motion speed of m will increase. If all collisions are perfectly elastic then maximum speed attained by m is (here $M \gg m$)



A) V_0 B) $\sqrt{\frac{m}{M}}v_0$ C) $\sqrt{\frac{M}{m}}v_0$ D) $2v_0$

21. A particle of mass m moving with a velocity v_0 enters a stationary object of mass M (free to move) as shown in the figure. All surfaces are frictionless and the system shown in the figure is lying on a smooth horizontal plane. Time taken by the particle of mass m to move from end A to end B of the object of mass M is



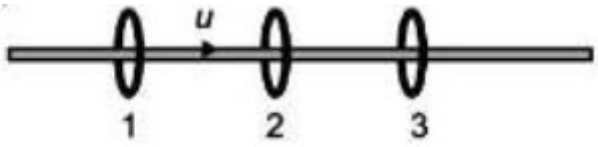
A) $\frac{2\ell+\pi R}{V_0}$ B) $\ell n\left(\frac{\ell+R}{v_0}\right)$ C) $\frac{\ell}{V_0}\ell n\left(\frac{\ell+\pi R}{\ell}\right)$ D) $\frac{\pi R}{v_0}\ell n\left(\frac{\ell+\pi R}{\pi R}\right)$

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
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Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

22. Three identical smooth steel beads are free to slide on a long fixed horizontal shaft. Beads 2 and 3 are at rest and are approached by the bead 1 at a speed u. The coefficient of restitution for all collisions is e , where $0 < e < 1$.



For More Join: @IITJEE_Advanced

A) Speed of bead 1 after the first collision is $\frac{u[1-e]}{2}$

B) Speed of bead 3 after the second collision is $\frac{u(1+e)^2}{4}$

C) Velocity of centre of mass of beads 2 and 3 after the second collision is $\frac{eu(1+e)}{4}$

D) Final velocity of centre of mass of beads 1,2 and 3 after the second collision is independent of the value of e

23. A metallic rod AB of length 1m is clamped at two points X and Y as shown in the figure such that $AX = BY = 0.3$ m.

Density and Young's modulus of the rod are $9.0\times10^3\,kgm^{-3}$ and $1.44\times10^{11}\,Nm^{-2}$ respectively. Standing longitudinal waves are set up in the rod. Then : (Assume clamped point as node)



A) Minimum frequency of oscillation of rod is 20 kHz

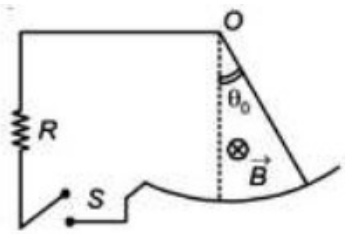
B) Minimum frequency of oscillation of rod is 10 kHz

C) Rod may oscillate with odd harmonics

D) Rod may oscillate with even harmonics

24. A conducting rod (mass M , length L) having uniform linear mass density is free to oscillate in a vertical plane about a horizontal axis passing through one of its end (O). The lower end of the rod slides along a smooth conducting, circular strip as shown in the figure.

A uniform magnetic field $\vec{B}$ exists in the region parallel to the axis of rotation in the direction as shown. If at $t = 0$, switch S is closed and rod is released from the position shown in diagram, then choose the correct alternatives (neglect effect of magnetic force on rod)



For More Join: @IITJEE_Advanced

A) Angular frequency of oscillations is $\sqrt{\frac{3g}{2L}}$

B) Maximum angular speed of rod will be $\theta_0\sqrt{\frac{3g}{2L}}$

C) Maximum induced current in the circuit is $\frac{BL^2\theta_0}{2R}\sqrt{\frac{3\,g}{2\,L}}$ if switch S is closed

D) Maximum induced current in the circuit is $\frac{BL^2\theta_0}{R}\sqrt{\frac{3\,g}{2\,L}}$ if switch S is closed

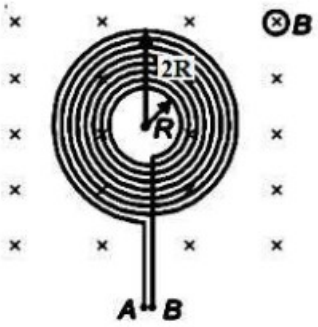
SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

25. A tube of length 1 m and radius a carries a viscous fluid whose velocity V is given as, $V = V_0\left(1-\frac{r^2}{a^2}\right)$ where r is the distance of fluid particle from the axis of the tube. The volume of fluid flowing across a section of tube in unit time is $\frac{\pi V_0 a^2}{n}$. Find the value of n.

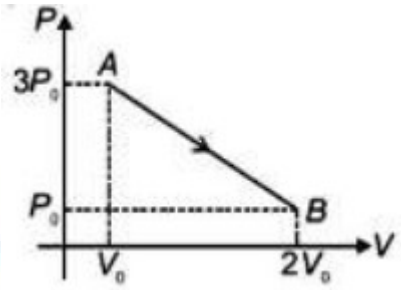
26. In a Coolidge tube the accelerating voltage is doubled and as a result the difference between the wavelength of K_α line and minimum X-ray wavelength is increased by 200%. If the wavelength of K_α line is $0.25\times P(\text{Å})$. Find the value of P. Initial accelerating voltage is 12.4 kV . (Given $hc = 12400\text{eV Å}$)

27. A planar spiral coil of inner radius R and outer radius $2R$ is made from a thin insulated wire which is tightly wound in N turns as shown in figure. A time varying magnetic field $B = B_0\sin\omega t$ is present in the region and it is directed perpendicular to the plane of the coil as shown. The amplitude of induced emf in the coil across the terminal A and B is given by $\frac{21\pi NR^2\omega B_0}{n}$. Find the value of n .

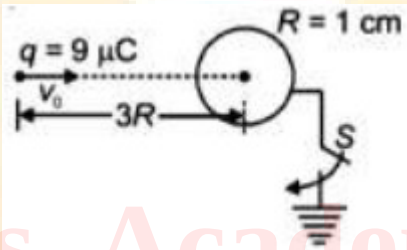


For More Join: @IITJEE_Advanced

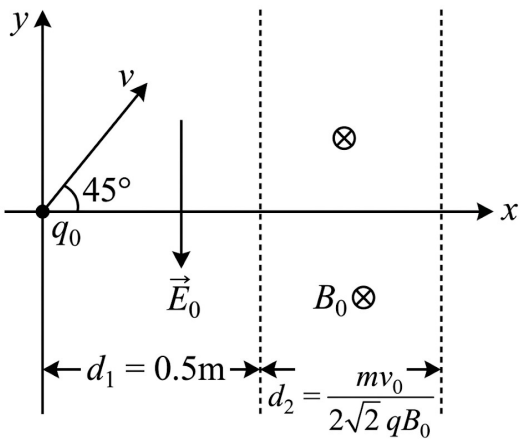
28. For the given thermodynamic process, 1 mole of an ideal gas is expanded from a volume V_0 to 2 V under the process AB . If the maximum temperature attained by the gas during the process is $T_{max} = \frac{nP_0 V_0}{8R}$, then find the value of n .



29. A point charge $q = 9\mu C$ is moving towards the center of conducting sphere which is earthed, at a speed of 100 m / s as shown. Find the current (in mA) flowing through the switch S at given instant.



30. A charged particle of charge ' q ' and mass ' m ' is projected in to a region which contains both the electric and magnetic fields as shown in the figure with velocity v at an angle 45° with x direction. If $v = \sqrt{\frac{qE}{m}}$, then find the net deviation in particle's motion (indegree). (Neglect the effect of gravity)

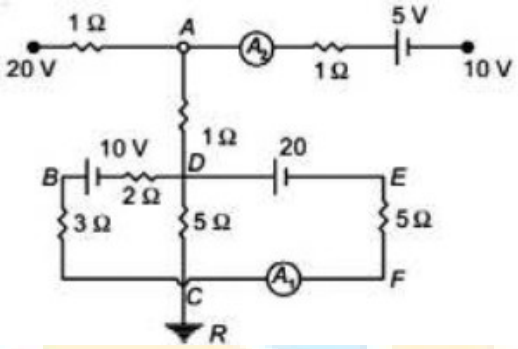


For More Join: @IITJEE_Advanced

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

31. In the adjacent circuit shown, all ammeters and batteries are ideal. Internal resistances of all the batteries are negligible. Point R is grounded. Match the List-I with List-II.

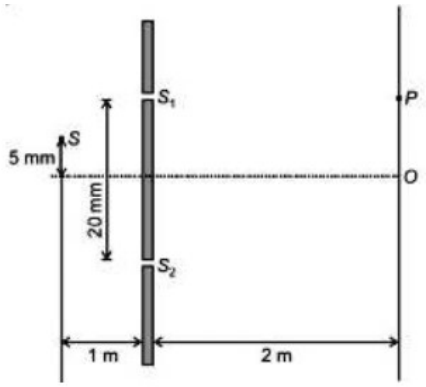


	List-I		List-II
P)	Potential at point B	1)	$\frac{15}{13}$ SI unit
Q)	Potential at point D	2)	$\frac{175}{13}$ SI unit
R)	Reading of Ammeter A_1	3)	3 SI unit
S)	Reading of Ammeter A_2	4)	$\frac{227}{13}$ SI unit
		5)	$\frac{18}{33}$ SI unit

A) $P \to 1; Q \to 2; R \to 5; S \to 4$
B) $P \to 4; Q \to 2; R \to 3; S \to 1$
C) $P \to 1; Q \to 3; R \to 2; S \to 4$
D) $P \to 4; Q \to 3; R \to 5; S \to 1$

For More Join: @IITJEE_Advanced

32. In Young's double-slit experiment, the point source S is placed 5 mm from the central axis as shown. If $\lambda = 500$ nm, then match the following list. Match List-I with List-II.



	List-I		List-II
P)	Nature and order of interference at point P, OP = 10 mm	1)	Bright fringe of order 200
Q)	Nature and order of interference at point O	2)	Bright fringe of order 380
R)	If a transparent paper having refractive index $\mu = 1.50$ and of thickness $t = 0.02$ mm is pasted on S_1. Then the nature and order of interference at P	3)	Bright fringe of order 400
S)	After inserting the transparent paper in front of the slit S_1, the nature and order of interference at 'O'	4)	Bright fringe of order 180
		5)	None of these

A) $P \to 1; Q \to 3; R \to 4; S \to 5$ B) $P \to 1; Q \to 3; R \to 2; S \to 4$

C) $P \to 3; Q \to 1; R \to 2; S \to 4$ D) $P \to 3; Q \to 2; R \to 1; S \to 5$

33. List-I specifies a point P at distance r from the centre (or) axis of symmetrical distribution of mass. List-II gives the variation of Gravitational field intensity at P as a function of $r(>0)$. Match the entries of List-I with List-II.

	List-I		List-II
P)	P lies outside a long cylinder having uniform mass density	1)	$g \propto \frac{1}{r^2}$
Q)	P lies inside a uniform spherical shell	2)	$g \propto \frac{1}{r}$
R)	P lies inside a uniform solid sphere	3)	$g \propto r$

For More Join: @IITJEE_Advanced

S)	P lies outside a solid sphere of radius R and its density varies as $\rho = \frac{P_0 x^2}{R^2}$, where x is distance from centre of the sphere	4)	$g \propto r^0$
		5)	$g \propto \frac{1}{r^3}$

A) $P \to 2; Q \to 4; R \to 3; S \to 1$ B) $P \to 4; Q \to 1; R \to 5; S \to 2$

C) $P \to 2; Q \to 3; R \to 2; S \to 4$ D) $P \to 4; Q \to 1; R \to 5; S \to 3$

34. A satellite of mass m is revolving in an elliptical orbit around a planet of mass M. When the satellite is at distance R from the planet, its velocity is $\sqrt{\frac{3}{2}\frac{GM}{R}}$ which is perpendicular to its acceleration. Let r, k, U and L be the distance, kinetic energy, potential energy and angular momentum of satellite at farthest point. Then match the entries in List-I with List-II.

	List-I		**List-II**
P)	$\frac{r}{R}$	1)	1
Q)	$\frac{2L^2}{3GMm^2R}$	2)	2
R)	$\frac{6kR}{GMm}$	3)	3
S)	$\left\|\frac{6UR}{GMm}\right\|$	4)	$\frac{1}{2}$
		5)	$\frac{1}{4}$

A) $P \to 3; Q \to 1; R \to 4; S \to 5$ B) $P \to 3; Q \to 1; R \to 4; S \to 2$

C) $P \to 3; Q \to 2; R \to 3; S \to 5$ D) $P \to 3; Q \to 2; R \to 4; S \to 1$

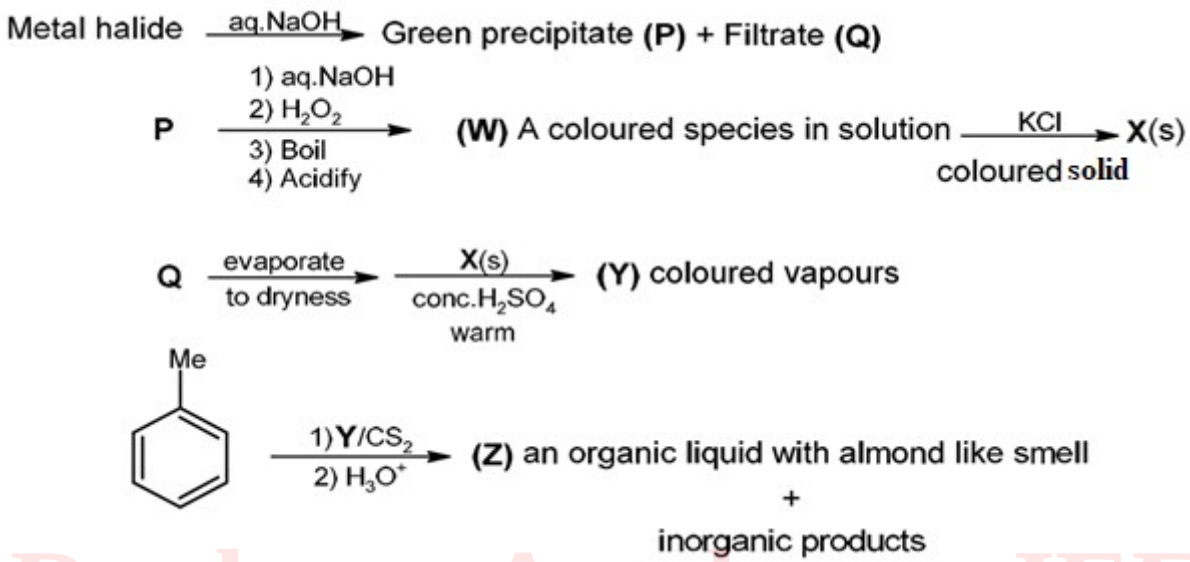
For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

35. In the scheme given below W, X, Y and Z respectively are



A) $Cr_2O_7^{2-}, K_2Cr_2O_7, CrO_2Cl_2$ and PhCOOH

B) $Cr_2O_4^{2-}, K_2Cr_2O_7, CrOCl_2$ and PhCHO

C) $FeO_4^{2-}, K_2FeO_4, FeCl_3$ and PhCHO

D) $Cr_2O_7^{2-}, K_2Cr_2O_7, CrO_2Cl_2$ and PhCHO

36. Plotting $\frac{1}{\Lambda_m}$ against $C\Lambda_m$ for aqueous solutions of a monobasic weak acid (HX) resulted in a straight line with Y -axis intercept of P and slope of S . The ratio $\frac{P^2}{S}$ is

[Λ_m = molar conductivity

Λ_m° = limiting molar conductivity

C = molar concentration K_a = dissociation constant of HX]

A) $K_a \wedge_m^\circ$ B) $\frac{1}{K_a}$ C) K_a D) $2 K_a \wedge_m^\circ$

For More Join: @IITJEE_Advanced

37. On decreasing the pH from 7 to 4 the solubility of a sparingly soluble salt AgY of a weak acid HY increases from 10^{-5} M to 10^{-4} M. Similarly, on decreasing the pH from 7 to 3 the solubility of a sparingly soluble salt AgZ of a weak acid HZ increases from 10^{-3} M to 10^{-2} M. Choose the incorrect statement.

A) The pH of 0.1 M HY is more than the pH of 0.1 M HZ .

B) The pH of 0.1 M NaY is more than the pH of 0.1 M NaZ

C) The difference between of $pK_a(HX)$ and a $pK_a(HY)$ is approximately 1.

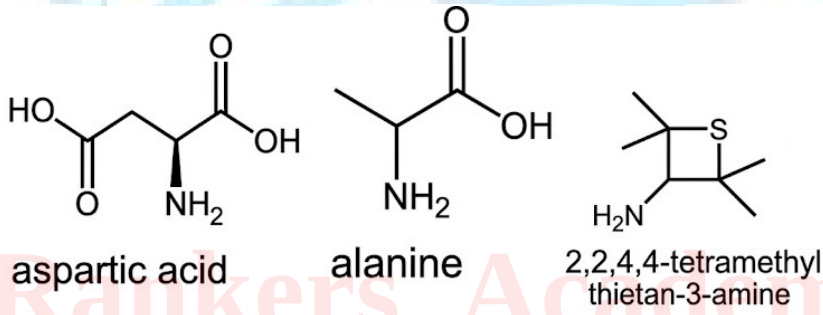
D) Limiting molar conductivity of HY must be more than that of HZ.

38. Alitame, an artificial sweetener 2000 times sweeter than sucrose is a dipeptide made up of the following components:

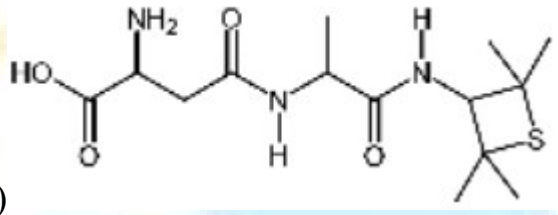
(i) Aspartic acid: An amino acid

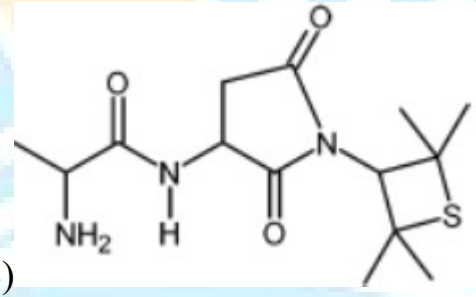
(ii) Alanine: An amino acid

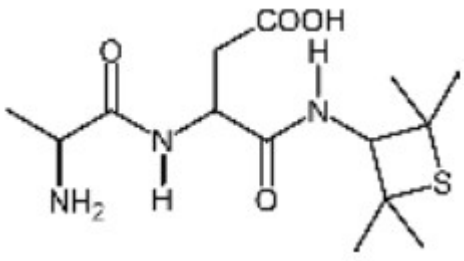
(iii) C-Terminal amide moiety: A novel amide moiety formed from 2, 2, 4, 4-tetramethylthietan-3amine

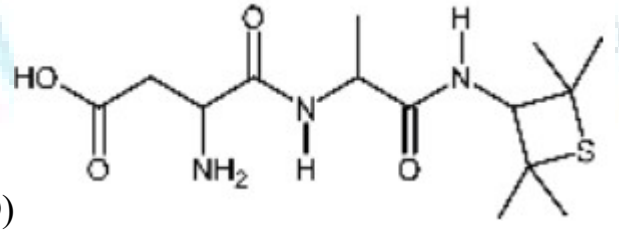


The structure of alitame is

A)


B)


C)


D)


For More Join: @IITJEE_Advanced

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Choose the correct statement(s):

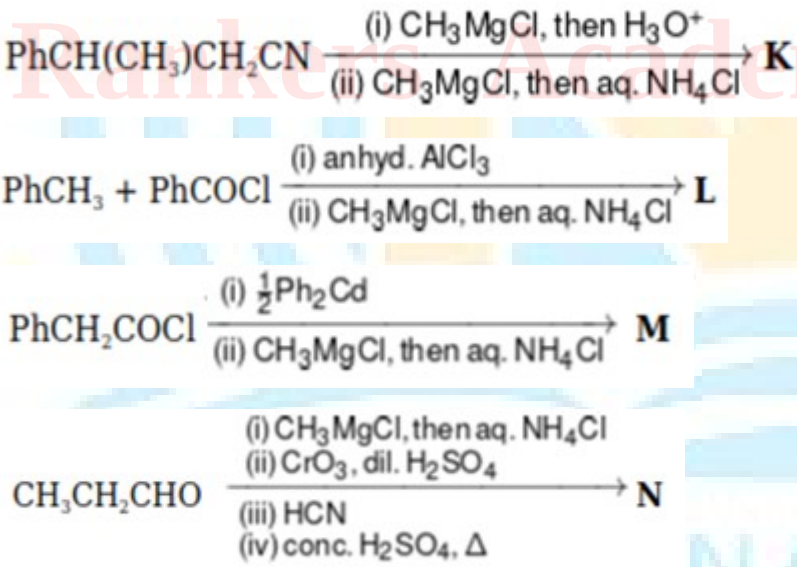
A) Kaolinite is a form of clay containing aluminium.

B) In froth floatation of sulphide ores NaCN forms soluble complex with PbS allowing separation of ZnS.

C) Bauxite ore usually contains silica, iron oxides and TiO_2 as impurities.

D) After concentration, mixture of oxide ores of iron is subjected to calcination/ roasting to remove water, to decompose carbonates and to oxidise sulphides.

40. In the following reactions, **K**, **L**, **M**, and **N** are the major products.



The correct statement(s) about **K**, **L**, **M**, and **N** is(are)

A) **L** has asymmetric carbon(s) and **N** has no asymmetric carbon(s).

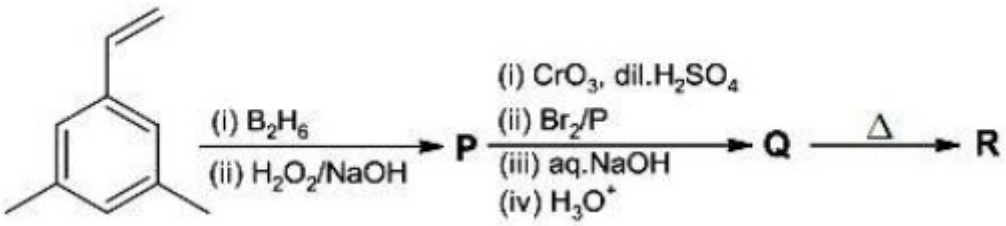
B) Both **K** and **N** have asymmetric carbon(s).

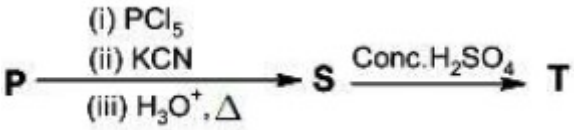
C) **L** and **M** are not isomers.

D) Both **L** and **M** have asymmetric carbon(s).

For More Join: @IITJEE_Advanced

41. Consider the following reaction scheme and choose the correct option(s) for the major products P, Q, R, S and T.





A) Both **P** and **Q** turn acidified dichromate green.

B) DU of **R** is 11 .

C) **S** gives effervescence with sodium bicarbonate.

D) **T** gives positive iodoform test.

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

42. Sodium dihydrogen phosphate on heating gives a dimer P which on further heating gives a hexameric sodium salt Q. Q finds application in removal of hardness of water. Assuming that the overall efficiency of conversion phosphorous in sodium dihydrogen phosphate to Q is 85%, the mass (in kg) of Q produced from 360 kg of sodium dihydrogen phosphate by the above process is

[Use, molar mass $\left(gmol^{-1}\right)$: $H=1, O=16, Na=23, P=31$]

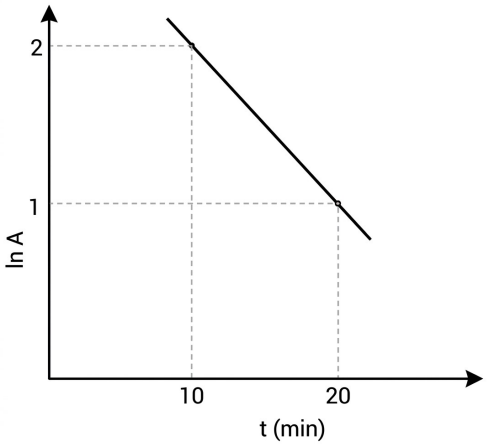
43. A gas has a compressibility factor of 0.75 and a molar volume of $0.6 dm^3\ mol^{-1}$ at a temperature of 600 K and pressure x atm . If it shows ideal gas behaviour at the same temperature and pressure, the molar volume will be $ydm^3\ mol^{-1}$. The value of x/y is ____ .

[Use: Gas constant, $R=8\times10^{-2}$ L atm mol^{-1} K^{-1}]

For More Join: @IITJEE_Advanced

44. Kinetics of radioactivity obeys first order kinetics, and the rate of radioactivity is independent of temperature. In a tracer study a radioactive isotope is injected into the subject. The following plot is obtained when $ln\,A$ is plotted against time (in minutes):

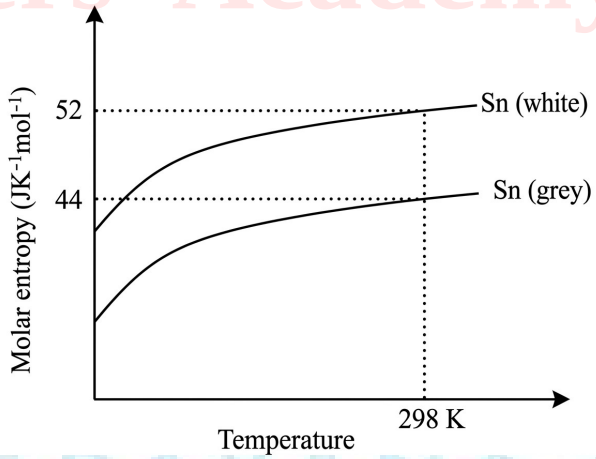
A - radioactivity in arbitrary units The value of $-ln\,A$ at 30 min. is _____



45. The effect of temperature on the absolute entropies of Sn (white) and Sn (grey) is given in the following graph. The Gibb's energy of formation of Sn (grey) at 298 K (in $Jmol^{-1}$)

is ____ .

Given, $\Delta_f H^\circ(Sn, grey) = -2.09 kJmol^{-1}$



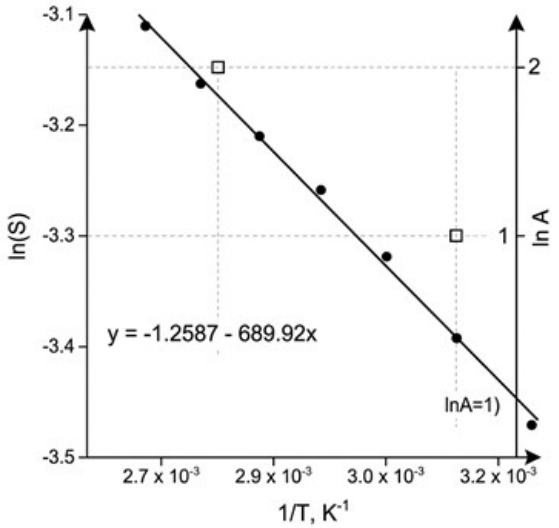
46. The plot of the natural logarithm of the molar solubility (S) of $Ag_2SO_4(s)$ as a function of the reciprocal of the absolute temperature is shown below $\Delta H^\circ r$ (in KJ/mol) for the following reaction :

$$Ag_2SO_4(s) \rightleftharpoons 2Ag^+(aq) + SO_4^{2-}(aq)$$

Given : $R = 8.3JK^{-1}\ mol^{-1}$

What is Answer to the nearest integer.

For More Join: @IITJEE_Advanced



47. Urea on heating gives a compound P by the following reaction:

$6NH_2CONH_2 \rightarrow 6NH_3 + 3CO_2 + P$

P is used in the manufacture of a polymer used to make unbreakable crockery. The sum of number of sp^2 hybridized carbon atoms, number of lone pairs and the number of electrons involved in pi bond formation is

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

48. Match the reactions (stoichiometry of the reactants not given) in List-I with one or more of their products given in List-II and choose the correct option.

	List-I		List-II
P)	$C_{12}H_{22}O_{11}$ + conc. $H_2SO_4 \rightarrow$	1)	H_2
Q)	Cu + conc. $H_2SO_4 \rightarrow$	2)	C
R)	S + conc. $H_2SO_4 \rightarrow$	3)	CO
S)	C + conc. $H_2SO_4 \rightarrow$	4)	CO_2
		5)	SO_2

A) $P \rightarrow 1,2; Q \rightarrow 1,5; R \rightarrow 5; S \rightarrow 4,5$ B) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 5; S \rightarrow 4,5$

C) $P \rightarrow 4,5; Q \rightarrow 5; R \rightarrow 1,5; S \rightarrow 1,4,5$ D) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 5; S \rightarrow 3,5$

For More Join: @IITJEE_Advanced

49. Match the complexes in List-I with one or more appropriate isomerism exhibited by them in List-II and choose the correct option.

	List-I		List-II
P)	$[Co(en)_2Cl_2]Br$	1)	Optical isomerism
Q)	$K_3[Cr(C_2O_4)_3]$	2)	Coordination isomerism
R)	$[Pt(NH_3)(H_2O)Cl_2]$	3)	Ionization isomerism
S)	$[Co(NH_3)_6][Cr(CN)_6]$	4)	Geometrical isomerism
		5)	Solvate isomerism

A) $P \rightarrow 1,3; Q \rightarrow 1; R \rightarrow 4,5; S \rightarrow 2$ B) $P \rightarrow 1,3,4; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$ D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

50. Match each reaction in List-I with the feature of its major product in List-II and choose the correct option.

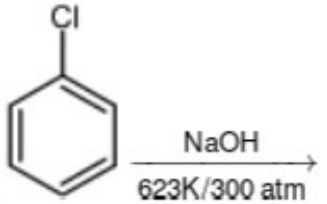
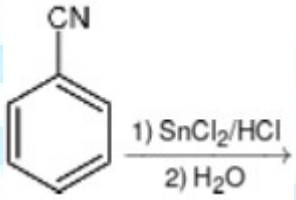
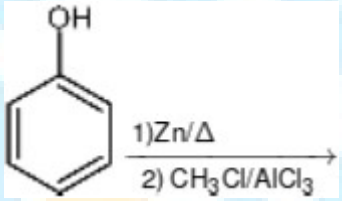
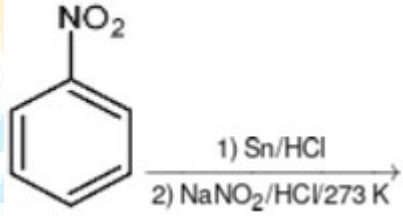
	List-I		List-II
P)	1-chloro-3-methylpentane $\xrightarrow[S_N2\ reaction]{aq.NaSH}$ (single stereoisomer)	1)	Inversion of configuration
Q)	2-bromobutane $\xrightarrow[S_N2\ reaction]{aq.NaSH}$ (single stereoisomer)	2)	Retention of configuration
R)	5-bromo-2,3-dimethylhexane $\xrightarrow[S_N1\ reaction]{aq.NaSH}$ (single stereoisomer)	3)	Mixture of enantiomers
S)	3-chloro-2-phenylpentane $\xrightarrow[S_N1\ reaction]{aq.NaSH}$ (single stereoisomer)	4)	Mixture of structural isomers
		5)	Mixture of diastereomers

A) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 3$ B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 4$

C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$ D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 3$

For More Join: @IITJEE_Advanced

51. The major products obtained from the reactions in List-II are the reactants for the named reactions mentioned in List-I. Match List-I with List-II and choose the correct option.

	List-I		List-II
P)	Cannizzaro reaction	1)	
Q)	Reimer-Tiemann reaction	2)	
R)	Sandmeyer reaction	3)	
S)	Etard reaction	4)	
		5)	

A) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 3$

B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$

C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 3$

D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

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